

Abraham Camelo Guerrero

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Professional Summary

Applied Machine Learning and Data Science professional with experience in ML experimentation, forecasting, NLP/LLM evaluation, data pipelines, and research-driven data products. Strong foundation in Engineering Physics and Information Systems, with a focus on translating complex data into practical insights. Collaborative and people-centered, committed to clear communication, team well-being, and responsible problem solving.

Education

York University – Master in Information Systems and Technology Sep. 2024 – Jun. 2026
Academic Excellence Fund.

National University of Colombia – B.Sc. in Engineering Physics Graduated Nov. 2024
Undergraduate tuition waiver awarded for top-10 academic GPA performance.

Technical Skills

- **Programming and Data Systems:** Python, SQL/MySQL, Git, Pandas, NumPy, Selenium, data pipelines, automation, reproducible workflows.
- **Machine Learning and Experimentation:** scikit-learn, model development, evaluation, benchmarking, backtesting, online learning, feature engineering.
- **Applied AI and Data Products:** LLMs, prompt engineering, embeddings, semantic similarity, NLP pipelines, Streamlit, Power BI, dashboards.

Experience

ML / Data Analysis Research Intern Dec. 2024 – Mar. 2026
VIBRAINT Toronto, ON, Canada

- Developed and evaluated machine learning pipelines for EEG-based predictive modeling, including model benchmarking, temporal validation, and performance analysis.
- Designed experiments for adaptive learning workflows, comparing offline, incremental, and rolling-window model update strategies.

Data Analyst Intern Jun. 2024 – Aug. 2024
MITACS @ York University Toronto, ON, Canada

- Built reproducible data pipelines for long-term environmental and crop phenology analysis, including data cleaning, interpolation, statistical analysis, and visualization.
- Analyzed temporal and geospatial patterns to support research insights, reporting, and stakeholder communication.

Data Analyst Intern Jul. 2023 – Jan. 2024
ISA Intercolombia Medellin, Colombia

- Supported analytics, automation, and business intelligence workflows using Python, SQL/MySQL, Power BI, and cloud-based machine learning tools.
- Developed dashboards, automated data processes, and improved computer vision model performance through dataset curation and model retraining.

Nanofabrication Laboratory Intern Apr. 2022 – Jun. 2022
King Abdullah University of Science and Technology Jeddah, Saudi Arabia

- Analyzed experimental semiconductor data using Python, MATLAB, Excel, and microscopy-based measurements to support technical reporting.

Applied Research Experience

Master's Researcher 2024 – 2026
York University Toronto, ON, Canada

- **LLM-based research summarization:** Built an NLP pipeline to evaluate LLM-generated summaries using

prompt engineering, embeddings, semantic similarity, and clustering-based analysis.

- **Time-series forecasting:** Developed a forecasting framework for seasonal signal analysis, decomposition, peak-period estimation, and multi-model evaluation.
- **Seasonal pattern detection:** Applied clustering, dimensionality reduction, and Gaussian Mixture Models to detect long-term seasonal shifts in climate data.
- **Research dashboard development:** Deployed an interactive Streamlit dashboard to communicate model outputs, forecasting results, and thesis findings.

Publications and Research Outputs

- Camelo Guerrero, A. I., Kozyrskiy, B., & Grachev, A. (2026). *Benchmarking label-revealed online updates for EEG BCI decoding*. In Proceedings of the Neuroadaptive Technology Conference 2026 (NAT'26). Berlin, Germany. Accepted and presented, in press.
- Camelo, A. I., & Khaiteer, P. A. (2026). *Investigating the Seasonality Factor with Gaussian Mixture Models*. In: Proceedings of International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2026), 5–7 February 2026, Boracay Island, Philippines, IEEE Xplore, pp. 1–6. DOI: 10.1109/ACDSA67686.2026.11467744.
- Camelo, A. I., & Khaiteer, P. A. (2026). *The Role of Remote Sensing in Desertification Research with a Case Study of La Guajira, Colombia*. In: M. Kumar et al. (eds.), Remote Sensing for Monitoring Sustainable Development, Springer Nature. In print.
- Camelo, A. I., Khaiteer, P. A., & Wang, Z. (2026). *Towards a Framework for Understanding Animal Behaviour and Climate Change: A Case Study of Little Penguins*. In: Information Technology and Systems, Springer Nature. Accepted.

Additional Experience

Teaching Assistant

- **Information Technology and Organizations @ York University:** Led tutorials, graded assignments, and supported student learning (Sep. 2024 – Aug. 2025).
- **Engineering Courses @ National University of Colombia:** Supported teaching and grading in Analog Electronics and ICT courses (Aug. 2021 – Jun. 2023).